MIL-C-50458 (MU)
24 September 1969
SUPERSEDING
PA-PD-2616B
27 November 1967

MILITARY SPECIFICATION

CASE, CARTRIDGE, COMBUSTIBLE, XM157E1, 152MM, PARTS FOR

1. SCOPE

1.1 This specification covers the component parts of one type of combustible cartridge case designated as Case, Cartridge, Combustible, 152MM, XM157E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-131	-	Barrier Material, Water-Vaporproof, Flexible
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-D-3464	-	Desiccants, Activated, Bagged; Packaging, Use and Static Dehumidification
MIL-I-45607	-	Inspection Equipment; Supply and Maintenance, For Ordnance
MIL-C-50269	-	Cellulose, Woodpulp (Sulfate) for XM157 and XM157E1 Combustible Cartridge Case

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellants, Solid; Sampling, Examination and Testing

FSC: 1320

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MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-20003 - Indicator Humidity Card, Three Spot, Impregnated Areas (Cobaltous Chloride)

FEDERAL

FED-STD-406 - Plastics; Method of Testing

DRAWINGS

ARMY

8799112 - Marking Diagram and Sealing of Ammunition Fiberboard Containers
8889489 - Box Fiber, for Body, for Case, Cartridge, Combustible, XM157E1
9225504 - Base, for Case, Cartridge Combustible, XM157E1
9227746 - Body, for Case, Cartridge, Combustible, XM157E1
9242503 - Box, Fiber, for Base, for Case, Cartridge, Combustible, XM157E1

PUBLICATIONS

ARMY

ET-9200660 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of inviation for bids shall apply.

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AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM-D-638	-	Tensile Properties of Plastics
ASTM-D-685	-	Conditioning Paper and Paper Products for Testing

(Copies of the above publications may be purchased directly from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania, 19103).

SPECIFICATION

NAVAL AIR SYSTEM COMMAND

XAS-1152	-	Film, Transparent, Flexible, Electrostatic-Free, Heavy Duty

(Copies of the above specification may be obtained upon request to the Naval Air Systems Command, Washington, D. C., 20360).

3. REQUIREMENTS

3.1 Materials.-The constituent materials shall comply with the requirements of all applicable drawings and specifications (See 6.5).

3.2 Parts.-The cartridge case parts shall comply with all the requirements specified in Drawings (dwgs.) 9225504 and 9227746, and with all the requirements specified in the applicable specifications.

3.3 Stability.-The nitrocellulose in the cartridge case body and base shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested in accordance with 4.3.3.1.

3.4 Physical properties

3.4.1 Tensile strength.-The cartridge case body and base shall have a minimum tensile strength of 700 psi when tested in accordance with 4.3.3.2.1.

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3.4.2 Weight.-The weight of the cartridge case body and base shall be determined in accordance with 4.3.3.2.2.

3.4.2.1 Body.-The XM157E1 cartridge case body shall weigh 190 plus or minus (±) 10 grams.

3.4.2.2 Base.-The XM157E1 cartridge case base shall weigh 57 ± 7 grams.

3.5 Candling test.-After molding, but prior to resin impregnation, the case body shall be free of imbedded foreign solids and visual evidence of molding defects. Examination shall be conducted in accordance with 4.3.3.3.

3.6 Profile-functional gage test.-The case body and base shall freely enter and position properly in a profile-functional gage. Testing shall be in accordance with 4.3.3.4.

3.7 Case strength.-The cartridge case body shall not crack, delaminate, or otherwise be permanently distorted when tested in accordance with 4.3.3.8.

3.8 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of molding flaws (such as delaminations and shredded edges), cracks, warpage, and foreign matter (See 6.15).

3.9 Proving ground.-The cartridge case, when assembled as part of a 152MM cartridge, shall function satisfactorily without excessive (See 6.7) or smoldering residue, when tested in accordance with 4.3.3.6.

3.10 Batch identity.-Until final acceptance, each case body and each case base shall be identified as to the batch mix from which it was fabricated (See 6.10).

3.11 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.12 Moisture content.-The moisture content of the case body and case base at time of final packaging shall be five (5) percent (%) maximum (max.), when tested in accordance with 4.3.3.7 (See 6.12).

3.13 Chemical analysis.-The case composition shall meet the requirements of the applicable part drawing when tested in accordance with 4.3.3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-When an inspection lot of hand-assembled cases (bodies and bases together) is submitted to the Government for acceptance, the contractor shall supply the following information, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the inspection lot of cases complies with all of the quality assurance provisions specified within this specification.

b. A statement that the components included in the inspection lot of cases (are) (are not) hybrid (See 6.14).

c. Number of components in the inspection lot that were inspected.

d. Number of components, identified by component lot numbers, in the inspection lot.

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e. A statement that the inspection lot (is) (is not) homogeneous as a result of b. or d., above (See 6.13).

f. Results obtained, by defect code, for all inspections performed.

g. Drawing, specification number, and date, together with identification and dates of changes.

h. Certificates of conformance.

i. Date submitted.

4.2 First article inspection

4.2.1 Examination and ballistic tests

4.2.1.1 Submission.-The contractor shall submit a first article sample, as designated by the contracting officer, for evaluation in accordance with the provisions of 4.2.1.2. The first article sample shall consist of 150 complete sets of parts (i.e., 150 cartridge case bodies and 150 cartridge case bases) which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures, molds, dies (See 6.8) and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government.)

4.2.1.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification, and any or all requirements of the applicable drawings. All tests shall be performed as specified in 4.4.

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4.2.2 Rejection.-If any component part fails to comply with the requirements for the inspections performed in accordance with 4.2.1.2, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample, or portion thereof. Until the initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the contracting officer.

4.2.3 New equipment approval.-Prior to the use of a new mold or press die in regular production of the case body or base, the contractor shall submit to the Government five (5) parts produced from the new mold or press die for inspection. If any part fails to comply with the applicable drawings or specification requirements, when tested as specified in 4.2.1.2, the mold or press die shall not be approved for use in regular production.

4.3 Inspection provisions

4.3.1 Lot formation

4.3.1.1 Case bodies and case bases.-The term "lot", as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous (See 6.13) collection of bodies or bases from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn, and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homoegeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions. Changes to the process, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with Standard MIL-STD-105. Inspection lots shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with Standard MIL-STD-1168. Each lot of case bodies or case bases shall contain:

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a. Nitrocellulose from one manufacturer, and from not more than two (2) production lots.

b. Diphenylamine from one manufacturer, and from one production lot.

c. Woodpulp from one manufacturer, and from one production lot.

d. Formvar resin from one manufacturer, and from one production lot (See 6.14).

4.3.1.2 Case, cartridge.-An inspection lot of cartridge cases shall consist of case bodies and case bases that are hand-assembled. Each case lot shall contain:

a. No more than one lot of case bodies, and

b. No more than one lot of case bases (See 6.13).

4.3.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105, except that inspection for critical defects, when tested, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed, using an AQL of 0.40 percent for each major defect, and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4. All examinations shall be conducted under standard atmospheric conditions, with samples that have been preconditioned for a minimum of 24 hours. Conditioning shall be in accordance with ASTM-D-685.

4.3.2.1 Body, for Case, Cartridge, Combustible, XM157E1 (See dwg. 9227746).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined.			
Major:	AQL 2.50 percent		
101.	Inside diameter of mouth, max.	See 4.4.2	01001
102.	Large O.D. at step	See 4.4.2	01002
103.	Small O.D. at step	See 4.4.2	01003

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104.	O.D. of straight cylindrical section	See 4.4.2	01004
105.	Distance from forward face to step	See 4.4.2	01005
106.	Distance from aft face to step	See 4.4.2	01006
Minor:	AQL 1.00 percent		
201.	Wall thickness	Gage	01007
202.	Workmanship (See 3.8)	Visual	01008

4.3.2.2 Base, for Case, Cartridge, Combustible, XM157E1 (See dwg. 9225504).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Inside diameter	See 4.4.2	02001
102.	Concentricity of primer cavity with inside diameter	See 4.4.2	02002
103.	Camber of base surface	See 4.4.2	02003
104.	Presence of flash hole	Visual	02004
Minor:	AQL 4.00 percent		
201.	Outside diameter, max.	See 4.4.2	02005
202.	Thickness	Gage	02006
203.	Diameter of boss	Gage	02007
204.	Diameter of flash hole	Gage	02008
205.	Depth of primer cavity	Gage	02009
206.	Diameter of primer cavity	Gage	02010
207.	Workmanship (See 3.8)	Visual	02011
208.	Presence of test hole	Visual	02012

4.3.2.3 Packaged case body or case base, prior to sealing of packing box.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 100 percent inspection		
101.	Protective packaging missing or torn	Visual	03001
102.	Heat seal incomplete on protective packaging	Visual	03002

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Minor: AQL 0.65 percent

201. Evidence of poor workmanship in assembly of protective packaging.....Visual 03003

4.3.3 Testing

4.3.3.1 Stability (See 3.3), Major defect (Body Code No. 01009; Base Code No. 02013).-A stability heat test at 134.5 degrees Centigrade (°C) shall be performed in accordance with Standard MIL-STD-286, Method 404.1.2. Random samples of five (5) case bodies and five (5) case bases shall be selected from each respective lot to provide the required test specimens. Failure of any sample is cause for lot rejection.

4.3.3.2 Physical properties

4.3.3.2.1 Tensile strength (See 3.4.1), Major defect (Body Code No. 01010; Base Code No. 02014).-The tensile breaking strength of the cartridge case body and base shall be determined in accordance with ASTM-D-638, using a cross head speed of one (1) inch per minute. Two test specimens of any convenient overall length, but with a one-inch gage length, one-inch gage width and thickness, in accordance with the applicable part drawing, shall be prepared from each of ten (10) randomly selected case bodies per lot. If one or more specimens fail to meet the applicable requirement in 3.4, the lot shall be rejected. Test specimens are to be conditioned for a minimum of 24 hours at 23° $\pm$ 2°C, at a relative humidity (R.H.) of 50 $\pm$ 5%, just prior to testing.

4.3.3.2.2 Weight (See 3.4.2), Major defect (Body Code No. 01011; Base Code No. 02015).-Samples for this test shall be selected in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235). All sample bodies and bases shall be weighed individually, using any Government-approved and suitably calibrated scale capable of reading to the nearest 0.5 gram (g). AQL for this defect shall be 0.40%. Test samples shall be preconditioned for a minimum of 24 hours prior to weighing at standard atmospheric conditions. Conditioning shall be in accordance with ASTM-D-685.

4.3.3.3 Candling test (See 3.5), Major defect (Body Code No. 01012).-The cartridge case body, after molding but prior to resin impregnation, shall be visually inspected to determine compliance with the applicable requirement. A high intensity, Government and safety approved light source shall be placed inside the body to aid inspection.

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A part shall be considered defective if any high or low density concentration or any solid foreign matter is evident, which has a dimension larger than 1/4 inch imbedded in the part. Sampling and inspection shall be in accordance with Standard MIL-STD-1235, CSP-1, Inspection Level II, with an AQL of 0.40 percent.

4.3.3.4 Functional gage test (See 3.6), Major defect (Body Code No. 01013; Base Code No. 02016).-Case bodies and bases shall be randomly selected from each lot in accordance with Standard MIL-STD-105, and subjected to a functional gage test. All testing shall be conducted at standard atmospheric conditions with samples that have been preconditioned for a minimum of 24 hours. Conditioning shall be in accordance with ASTM-D-685. Testing shall be in accordance with 4.4.2. The AQL for this test shall be 0.40 percent.

4.3.3.5 Composition (See dwgs. 9225504 and 9227746), Major defect (Body Code No. 01014; Base Code No. 02017), (See 3.13).-From each lot, five (5) case bodies and five (5) case bases, fabricated from separate batch mixes (See 6.10) shall be selected for chemical analysis in accordance with 4.4.1. Duplicate specimens from each sample body and base shall be analyzed. If the average of any set of duplicate analyses fails to comply with the applicable requirement, the lot shall be rejected.

4.3.3.6 Proving ground (See 3.9), Smoldering residue/Special defect, Code No. 04001; Excessive residue/Major defect, Code No. 04002.-One-hundred twenty (120) cartridge cases (See 4.3.1.2) shall be randomly selected from each lot until three (3) consecutive lots have been accepted; after three(3) lots have been accepted, forty (40) case bodies and forty (40) case bases shall be randomly selected for proving ground testing. Sample cases shall be assembled to 152MM projectiles and ballistically tested as specified in 4.4.4. Failure to comply with the applicable requirement when tested at 70° Fahrenheit (F) shall be cause for lot rejection. Firings at -40°F and +125°F shall be for information only (See 6.11).

4.3.3.7 Moisture content (See 3.12), Major defect (Body Code No. 01015; Base Code No. 02018).-Just prior to final packaging, the moisture content of case bodies and bases shall be determined in accordance with 4.4.5 and CSP-1, Level II; Continuous Sampling Plan, per MIL-STD-1235.
TD-1235.

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4.3.3.8 Case strength (See 3.7), Major defect, Code No. 01016.-Samples shall be randomly selected from each case body lot in accordance with Standard MIL-STD-105 General Inspection Level II. The AQL for this defect is 0.15 percent. Testing shall be conducted as specified in 4.4.3. Any cracking, delamination, or permanent distortion of any sample case body shall be cause for rejection of the body lot, and of any bases produced from the same batch(es) mix as the rejected bodies.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9200660 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Composition

4.4.1.1 Preparation of sample

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4.4.1.1.1 Body (See 4.3.3.5).-From each sample case body, two (2) specimens of approximately one (1) square inch shall be taken from the sidewall (one from the base end of the sidewall and one approximately 3 1/2 inches from the base end). Each specimen shall be divided into 1/8-inch (approximate) cubes or equivalent. The cubes shall then be thoroughly mixed to form a composite. From the composite, two (2) one-gram specimens shall be selected, dried for three hours at 100° ± 5°C and weighed to the nearest 0.2 milligram (mg). This weight is designated as the dry weight of the prepared sample (W). This procedure shall be repeated for each sample body.

4.4.1.1.2 Base (See 4.3.3.5).-From each sample base, two (2) specimens of approximately one (1) square inch shall be cut from opposite sections of the base. A composite shall then be formed, dried and weighed as specified in 4.4.1.1.1. This procedure shall be repeated for each sample base.

4.4.1.2 Chemical analysis

4.4.1.2.1 Extraction.-Using a tared, extra-coarse, sintered glass thimble, extract each sample prepared in accordance with 4.4.1.1, with methylene chloride, for a minimum of fourteen (14) hours with a Soxhlet extractor, or six (6) hours with a Roweg extractor. Evaporate the extracts to dryness, using a current of clean, dry air. This extract is to be used for the Diphenylamine determination in 4.4.1.2.3. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernable. The thimble plus residue should be dried for three hours at 100° ± 5°C. Determine the dry weight of this residue (E) to the nearest 0.2 mg. This residue is to be used for the Nitrocellulose determination in 4.4.1.2.2. Percent extractables shall be calculated by the following equation:

$$\text{Percent extractables} = \frac{(W - E) \times 100}{W}$$

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4.4.1.2.2 Nitrocellulose determination.-From the dried residue obtained in 4.4.1.2.1, weigh out to the nearest 0.2 mg a specimen of 0.3 to 0.4 g (Weight I). Place this specimen in a 500 milliliter (ml) flask with a flat bottom, a side arm for the admittance of inert gas, and fitted with a standard taper ground-glass joint to match the condenser to be used during refluxing), and add to each flask 25 ml of n-butyl acetate and 45 ml of glacial acetic acid. Reflux each specimen for five (5) minutes with a stream of carbon dioxide passing continuously into the flask. Add to the flask, via the condenser, 25 ml of ferrous ammonium sulphate 0.7N (prepared as per Standard MIL-STD-286, Method 604.1.1, except that the volume used for the blank determination shall be 25 ml in lieu of 15 ml), followed by 10 ml of concentrated hydrochloric acid. Refluxing is then continued for thirty (30) minutes to give a clear, lemon yellow liquid, in which the specimen will have completely disintegrated. With the inert gas still running, and the condenser still attached to the flask, cool in a water bath to room temperature. The flask is then removed from the condenser for titration and, under a stream of inert gas, 5 ml of 20% ammonium thiocyanate (Standard MIL-STD-286, Method 706.1) are added and then titrated with a 0.2N titanous chloride solution (Standard MIL-STD-286, Method 601.1). A blank titration on the reagents is run, and the percent nitrocellulose calculated as follows:

$$\text{Percent Nitrocellulose} = \frac{46.69\ (V - B)\ N\ E \times 100}{CWI}$$

Where:

V = Volume of titanous chloride used for the specimen

B = Volume of titanous chloride used for blank

I = Weight of specimen

N = Normality of titanous chloride solution

E = Weight of residue

W = Weight of prepared sample

C = Nitrogen content of nitrocellulose

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4.4.1.2.3 Diphenylamine determination.-To the dried extract from 4.4.1.2.1, add 50 ml of glacial acetic acid and, if necessary, warm slightly to dissolve the extract; then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1 (starting with 5.5) of Standard MIL-STD-286, except that 25 ml of methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color) the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further blue color. In the final calculation, weight (W) is the value determined in 4.4.1.1 above, and designated as the weight of the prepared sample.

4.4.1.2.4 Woodpulp.-The percent woodpulp contained in the composite sample shall be calculated from the values obtained in 4.4.1.2.2, using the following equation:

$$\text{Percent woodpulp} = \frac{E}{W}\left[100 - \left(\frac{3.705\ (V - B)\ 100\ N}{I}\right)\right]$$

Where:

E = Dry weight of residue

W = Dry weight of prepared composite sample

V, B and N are as specified in 4.4.1.2.2.

4.4.1.2.5 Resin and additive determination.-The percent resin plus additives present in the composite is calculated by the following formula:

Percent resin + additives = 100 - Percent (Nitrocellulose, DPA and Woodpulp) as determined in 4.4.1.2.2 through 4.4.1.2.4.

4.4.2 Functional gage testing.-This test shall be conducted in accordance with the instructions contained on the applicable Inspection Equipment Drawing (See 4.3.4). Observation shall be made for compliance with the requirement of 3.6.

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4.4.3 Case strength.-Testing shall be conducted in accordance with the operating instructions contained on the applicable Inspection Equipment Drawing (See 4.3.4). Observation shall be made for compliance with the requirement of 3.7 (See 6.6).

4.4.4 Proving ground.-Testing shall be in accordance with the applicable Aberdeen Proving Ground Acceptance Test Procedure. Testing shall be at a Government Proving Ground designated by the contracting officer. The test samples shall be fired at the temperatures indicated in Table I, and accepted in accordance with Table II.

TABLE I

Temperature	First Article	First 3 Lots	Subsequent Lots
- 40°F	20	20	--
+ 70°F	80	80	40
+125°F	20	20	--

TABLE II*

Sample Size	Acceptance Criteria for Smoldering Residue		Acceptance Criteria for Excessive Residue	
	Accept	Reject	Accept	Reject
80	0	1	1	2
40	0	1	1	2

*All test firings at -40°F and +125°F shall be for information only.

Observation shall be made for compliance with the applicable requirement, 3.9. Camera coverage of the breech block and chamber of the weapon shall be provided for all test firings.

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4.4.4.1 Refiring.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the results shall be declared invalid, and a new test shall be performed.

4.4.5 Moisture content.-The moisture content of the case body and case base shall be determined using test equipment in accordance with 4.3.4 and applicable manufacturer's operating instructions. One reading shall be taken on the cylindrical portion of the case body and one reading shall be taken on the case base (See 6.12).

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level C.-Each case body shall be sealed in a 4 mil (minimum thickness) polyolefin bag. A maximum of twenty (20) case bases shall be packed together in a similar polyolefin bag. The material shall be black with surface resistivity not greater than 10^3 ohms per centimeter, when determined in accordance with Standard FED-STD-406, Method 4041 and 4052. The electrostatic bleed-off shall comply with the requirements of Specification XAS-1152, paragraph 4.7.10, and Table I. The method of sealing shall be a tape-secured fold or Government-approved heat seal.

5.2 Packing

5.2.1 Level C.-Bagged case bodies and case bases shall be packed in accordance with dwg. 8889489 or 9242503, as applicable.

5.3 Marking.-Unless otherwise specified, markings shall be in accordance with dwg. 8799112.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each case inspection lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

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6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection Equipment Lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment designs or codes, as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment, as defined in 6.3.3 and 6.3.4, will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of first article sample.-Instructions as to the location for the evaluation of first article samples shall be obtained from the contracting officer, upon request. All inquiries should be forwarded to:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NK1
Dover, New Jersey 07801

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6.5 Additives and solvents.-All additives (i.e., mold release agents and process aids) and solvents (i.e., isopropyl alcohol, toluene, etc.) shall have Government approval prior to their use in production.

6.6 Case strength test.-To reduce distortion of the case body during case strength testing, the cartridge case base or other equivalent removable cap is permissible as support for the case body.

6.7 Excessive residue.-Residue is defined as excessive if it is significant enough to prevent chambering or hinder functioning of a subsequent round, as determined by the proving ground.

6.8 First article sample.-The first article sample shall contain a representative quantity (a minimum of five [5]) of bodies and bases fabricated from each felting mold, molding die, and restrike die intended for use in regular production. Each body and base sample shall be appropriately labelled as to felting mold, molding die, and restrike mold used during its fabrication.

6.9 Packaging.-To minimize the possibility of residual solvent being trapped in the sealed package, a minimum waiting period of 24 hours at 70° ± 5°F or one hour at 125 ± 5°F shall be observed from time of fabrication to time of packaging.

6.10 Batch mix.-A batch mix is defined as an aqueous mixture of nitrocellulose, woodpulp, and approved additives (in the supply tank) to which no material additions except water have been made since the initial production part was fabricated from the batch.

6.11 Proving ground

6.11.1 Assembly of cartridge case.-Cartridge cases shall be assembled to 152MM projectiles using normal loading procedures, except that no epoxy shall be used at the case-projectile joint, and no nitrocellulose lacquer shall be used to cement the case body to the case base, or in the assembly of the XM91E1 Primer.

6.11.2 Cleaning.-Prior to each firing, the breech and firing probe of the test weapon shall be thoroughly cleaned and dried.

6.11.3 Handling and storage.-Special care shall be taken in the handling and storage of sample cartridge cases to prevent contamination by grease, grime, dirt, or other foreign material.

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6.12 Rework.-In the event that a case part fails to comply with the moisture requirement, it may be dried, re-inspected, and returned to the lot. Drying shall be accomplished only by a Government- and safety-approved method.

6.13 Homogeneous lots.-A component lot shall be considered homogeneous if the lot contains source materials in compliance with 4.3.1.1; i.e., contains no more than two (2) lots of nitrocellulose, one (1) lot of woodpulp, one (1) lot of Formvar resin, and one (1) lot of DPA. An inspection lot of hand-assembled cases shall be considered homogeneous if it contains only one (1) homogeneous body lot and one (1) homogeneous base lot. The body and base components are not required to be of the same component lot designation or of the same lot number as the case lot. However, only one (1) body and one (1) base lot is permissible in a case lot as defined in 4.3.1.2.

6.14 Hybrid lots.-A component lot shall be considered hybrid if the lot contains source materials which are not in compliance with 4.3.1.1; ie, the lot contains more than two (2) lots of NC, one (1) lot of Formvar resin, woodpulp, or DPA. An inspection lot of hand-assembled cartridge cases shall be considered hybrid if: (1) the lot contains a body lot or base lot that is hybrid and/or (2) the lot contains more than one (1) body lot or base lot. In either condition of (1) or (2), the entire case lot shall be considered hybrid.

6.15 Warpage.-Warpage is permitted within the envelope tolerance, but not outside of envelope tolerance.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1320-A-163

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Picatinny Arsenal
Dover, New Jersey 07801

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